

Module 7: Mini Project-2

Patient Survival Prediction using XGBoost ML Monitoring with Prometheus & Grafana

For this project, you will be required to create a real-time dashboard and alerts for monitoring Patient Survival Prediction application.

In the previous mini-project session, you have already trained a XGBoost model to predict survival of patients with heart failure. After that, you saved the trained model, and created a Gradio-based app. You can use the existing application files from the previous mini-project session.

Please refer to the “*Demo session - Monitoring Tools*” held on April 6th during Module-6 for this mini-project.

Step 1: Run your application on GitHub Codespaces OR Hugging Face Spaces [2 point]

- 1.1 Take the testing dataset from Colab and save into a csv file
- 1.2 Update your application files to include prometheus-client library and methods to send prometheus supported metrics whenever a request is made to **/metrics** endpoint
- 1.3 Create a Dockerfile to dockerize the application
- 1.4 Run your application either on GitHub Codespaces OR using Hugging Face Spaces
- 1.5 Access your application and check its **/metrics** endpoint (Debug if error persists)

Step 2: Setup Prometheus & Grafana using GitHub Codespaces [2 point]

- 2.1 Create *prometheus.yml* configuration file and include your application as a target
- 2.2 Start Prometheus using docker on Codespaces
- 2.3 Start Grafana Open Source using docker on Codespaces

Step 3: Access the application's metrics in Prometheus [1 point]

- 3.1 Access your Prometheus service on the port forwarded address

- 3.2 Go to the Targets page and check the State of the target application
- 3.3 On Graph tab, query your application metric

Step 4: Create Real-time Dashboard in Grafana [2 points]

- 4.1 Create a New Dashboard in Grafana
- 4.2 Add visualization panels to it to monitor the metrics coming from Prometheus service
- 4.3 Update panel settings such as title, description, min & max values, threshold, etc
- 4.4 Save your Dashboard

Step 5: Add an Alert in Grafana [2 points]

- 5.1 Add a new Alert rule in Grafana (Make sure to link it with a panel in your dashboard)
- 5.2 Add a new Contact point such as Email
- 5.3 Add a new Notification policy
- 5.4 Check if the notification is being sent to the contact point when the alert rule is in breach (You can alter the threshold to let alert rule in breach)